SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING

SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE

Exp 4: Write the python program for Cript-Arithmetic problem

CODE:

```
from itertools import permutations

letters = "SENDMORY"
digits = range(10)

for p in permutations(digits, len(letters)):
    d = dict(zip(letters, p))

    if d["S"] == 0 or d["M"] == 0:
        continue

    SEND = (d["S"] * 1000 + d["E"] * 100 +
            d["N"] * 10 + d["D"])

    MORE = (d["M"] * 1000 + d["O"] * 100 +
            d["R"] * 10 + d["E"])

    MONEY = (d["M"] * 10000 + d["O"] * 1000 +
             d["N"] * 100 + d["E"] * 10 + d["Y"])

    if SEND + MORE == MONEY:
        print("SEND =", SEND)
```

```
print("MORE =", MORE)
print("MONEY =", MONEY)
break
```

OUTPUT:

